

Note set #5

Fields in terms of Potentials Read Chapter 6.1-6.7 Balanis

Assume homogeneous space of infinite extent (μ, ε) in which electric but not magnetic sources reside.

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} \quad (1)$$

$$\nabla \times \mathbf{H} = j\omega\varepsilon\mathbf{E} + \mathbf{J} \quad (2)$$

$$\nabla \cdot \mathbf{E} = \rho / \varepsilon \quad (3)$$

$$\nabla \cdot \mathbf{H} = 0 \quad (4)$$

produce the general equation, Notes pg. 1-6,

$$\nabla^2 \mathbf{E} + k_c^2 \mathbf{E} = j\omega\mu\mathbf{J}^i - \frac{\nabla(\nabla \cdot \mathbf{J}^i)}{\sigma + j\omega\varepsilon}$$

which could not be solved.

Compare $\nabla \cdot \mathbf{H} = 0$ with the identity Appendix (II-40) Balanis

$$\nabla \cdot \nabla \times \mathbf{A} = 0$$

which is true for any $\mathbf{A}$.

$$\Rightarrow \boxed{\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A}} \quad (5)$$

$\mathbf{A} \triangleq$ magnetic vector potential (Amps). Substitute (5) into (1)

$$\nabla \times \mathbf{E} = -j\omega \nabla \times \mathbf{A}$$

or

$$\nabla \times (\underbrace{\mathbf{E} + j\omega\mu\mathbf{A}}) = 0$$

set $-\nabla\Phi$ using the identity $\nabla \times \nabla\Phi = 0$, true for any scalar function Φ (II-41)

$$\Rightarrow \boxed{\mathbf{E} = -j\omega\mathbf{A} - \nabla\Phi} \quad (6)$$

where $\Phi \equiv$ scalar potential (volts)

Substitute (5) and (6) into (2)

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E} + \mathbf{J} \quad (2)$$

$$\longrightarrow \nabla \times (\nabla \times \mathbf{A}) = j\omega\epsilon\mu(-j\omega\mathbf{A} - \nabla\Phi) + \mu\mathbf{J}$$

$$\begin{aligned} \underbrace{\nabla \times \nabla \times \mathbf{A}}_{\text{identity A-14}} &= \mu\mathbf{J} + \omega^2\mu\epsilon\mathbf{A} - j\omega\epsilon\mu\nabla\Phi \\ -\nabla^2\mathbf{A} + \nabla\nabla \cdot \mathbf{A} &= \mu\mathbf{J} + \omega^2\mu\epsilon\mathbf{A} - j\omega\epsilon\mu\nabla\Phi \\ -\nabla^2\mathbf{A} - \omega^2\mu\epsilon\mathbf{A} &= \mu\mathbf{J} - \underbrace{\nabla(\nabla \cdot \mathbf{A} + j\omega\mu\epsilon\Phi)}_{?} \\ &\doteq k^2 \end{aligned}$$

We can choose $\nabla \cdot \mathbf{A}$ because Helmholtz theorem states that a vector is specified by its divergence and curl. We created the vector $\mathbf{A}$ by defining its curl. In this step we fix its divergence.

$$\Rightarrow \nabla \cdot \mathbf{A} = -j\omega\epsilon\mu\Phi \quad (7) \quad \underline{\text{Lorenz gauge condition}}$$

$$\boxed{\therefore \nabla^2\mathbf{A} + k^2\mathbf{A} = -\mu\mathbf{J}} \quad (8)$$

Note (7) can be written $\Phi = -\frac{1}{j\omega\mu\epsilon}\nabla \cdot \mathbf{A}$ which can be substituted into (6) to give

$$\mathbf{E} = -j\omega\mathbf{A} - \frac{j}{\omega\epsilon\mu}\nabla\nabla \cdot \mathbf{A}$$

$$\text{or} \quad \boxed{\mathbf{E} = -\frac{j\omega}{k^2}(k^2\mathbf{A} + \nabla\nabla \cdot \mathbf{A})} \quad (9)$$

Also by taking the divergence of $\mathbf{E} = -j\omega\mathbf{A} - \nabla\Phi$ (6)

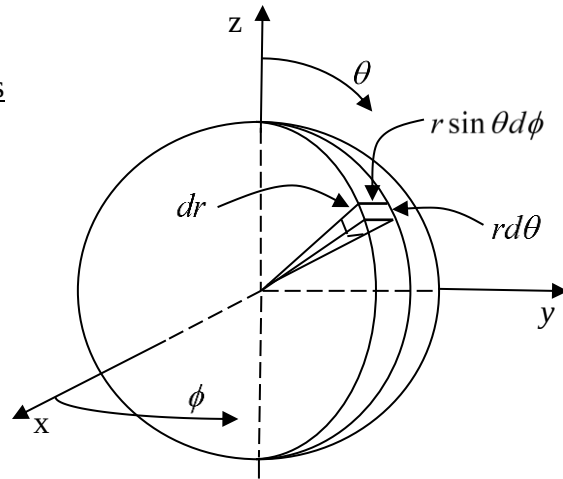
$$\begin{aligned} \nabla \cdot \mathbf{E} &= -j\omega\nabla \cdot \mathbf{A} - \underbrace{\nabla \cdot \nabla\Phi}_{=\nabla^2\Phi} \\ &\stackrel{\substack{=\rho/\epsilon \\ \text{from (3)}}}{=} \quad \quad \quad \nearrow \quad \quad \quad (7) \end{aligned}$$

we get

$$\boxed{\nabla^2\Phi + k^2\Phi = -\rho/\epsilon} \quad (10)$$

Differential Surface and volume elements

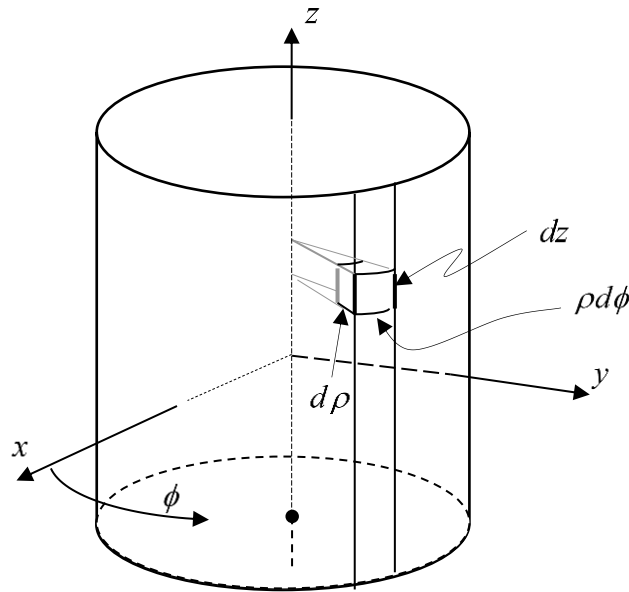
- Spherical coordinates (r, θ, ϕ)



$$d\mathbf{s} = r^2 \sin \theta d\theta d\phi \hat{\mathbf{r}} \text{ differential surface}$$

$$dv = r^2 \sin \theta dr d\theta d\phi \text{ -differential volume}$$

- Cylindrical coordinates (ρ, ϕ, z)



$$d\mathbf{s} = \rho d\phi dz \hat{\mathbf{\rho}} \text{ differential surface element of the cylinder}$$

$$dv = \rho d\rho d\phi dz \text{ - differential volume of the cylinder}$$

δ - function

$$\int_a^b \delta(x-c) dx = 1 \quad ; \quad a \leq c \leq b$$

$$\int_a^b \delta(x-c) f(x) dx = f(c) \quad , \quad a \leq c \leq b$$

- Cylindrical coordinates: $dv = \rho d\phi d\rho dz$

$$\int_a^b \delta(\rho - \rho') d\rho = 1 \quad ; \quad a \leq \rho' \leq b$$

$$\int_a^b \left[\frac{\delta(\phi - \phi')}{\rho} \right] \rho d\phi = 1 \quad ; \quad a \leq \phi' \leq b$$

$$\int_a^b \delta(z - z') dz = 1 \quad ; \quad a \leq z' \leq b$$

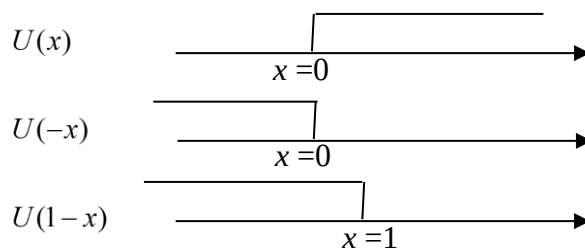
- Spherical coordinates: $dv = r^2 \sin \theta dr d\theta d\phi$
 $= \underbrace{r \sin \theta d\phi}_{\text{area element}} r d\theta dr$

$$\int_a^b \left[\frac{\delta(\phi - \phi')}{r \sin \theta} \right] r \sin \theta d\phi = 1 \quad ; \quad a \leq \phi' \leq b$$

$$\int_a^b \left[\frac{\delta(\theta - \theta')}{r} \right] r d\theta = 1 \quad ; \quad a \leq \theta' \leq b$$

$$\int_a^b \delta(r - r') dr = 1 \quad ; \quad a \leq r' \leq b$$

Unit Step Function: Zero except when the argument is positive



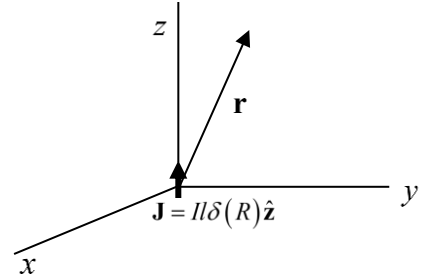
Vector potential for a point source

In general we must solve the magnetic vector potential eqn.

$$\nabla^2 \mathbf{A} + k^2 \mathbf{A} = -\mu \mathbf{J} \quad (1)$$

Now assume the current is a point source polarized in the z –direction and having an electric moment Il

$$\mathbf{J} = Il\delta(R)\hat{\mathbf{z}} = Il\delta(r)\frac{\delta(\phi)}{r\sin\theta}\frac{\delta(\theta)}{r}\hat{\mathbf{z}} \quad (2)$$



(1) becomes

$$\nabla^2 A_z + k^2 A_z = -\mu Il\delta(R) \quad (3)$$

Because the current is a point source, according to eqn. (3) the vector potential should be spherically symmetric (i.e. $A_z = A_z(r)$ and is not a function of θ and ϕ). Therefore (3) reduces to

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} A_z(r) \right) + k^2 A_z(r) = -\mu Il\delta(R) \quad (4)$$

One way to solve this equation is to first solve the homogeneous equation

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} A_z(r) \right) + k^2 A_z(r) = 0 \quad (5)$$

in terms of unknown coefficients and then find the coefficients by enforcing eqn. (4). We will use this method – which is known as Green's method.

Clever trick: Notice that if you let

$$A_z(r) = \frac{A'_z(r)}{r} \quad (6)$$

in $\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} A_z(r) \right) + k^2 A_z(r) = 0$ (5). The result is

$$\frac{d^2 A'_z}{dr^2} + k^2 A'_z = 0 \quad (7)$$

which has the solution

$$A'_z = C_1 e^{-jkr} + C_2 e^{jkr} \quad (8)$$

Substitute (8) into (6) to get

$$A_z = C_1 \frac{e^{-jkr}}{r} + C_2 \frac{e^{jkr}}{r} \quad (9)$$

Eqn. (9) must satisfy two boundary conditions:

- 1st boundary condition - the wave must be outward traveling from the source and decay to zero at ∞ (radiation boundary condition).

If the wave must be outward traveling from the source then e^{jkr} cannot be a solution, so $C_2 = 0$ and (9) becomes

$$A_z = C_1 \frac{e^{-jkr}}{r} \quad (10)$$

The $1/r$ behavior of the solution (9) provides the necessary decay at infinity.

- 2nd boundary condition – the field (10) must satisfy (4) as $r \rightarrow 0$ (i.e. at the origin)

To satisfy this condition we integrate (4) over a spherical region that has a radius r_a and let $r_a \rightarrow 0$

$$\lim_{r_a \rightarrow 0} \left\{ \int_{\nu} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} A_z(r) \right) d\nu + k^2 \int_{\nu} A_z(r) d\nu = -\mu I l \int_{\nu} \delta(R) d\nu \right\} \quad (11)$$

- The 3rd term becomes

$$-\mu Il \int_V \delta(R) dV = -\mu Il \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^{r_a} \delta(r) \frac{\delta(\phi)}{r \sin \theta} \frac{\delta(\theta)}{r} r^2 \sin \theta dr d\theta d\phi = -\mu Il \quad (12)$$

- The 2nd term becomes

$$\begin{aligned} \lim_{r_a \rightarrow 0} k^2 \int_V A_z(r) dV &= \lim_{r_a \rightarrow 0} k^2 \int_V C_1 \frac{e^{-jkr}}{r} r^2 \sin \theta dr d\theta d\phi \\ &= \lim_{r_a \rightarrow 0} k^2 C_1 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \frac{r_a^2}{2} \sin \theta d\theta d\phi \rightarrow 0 \end{aligned}$$

(13)

- The 1st term is evaluated as follows:

Recall that

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} A_z(r) \right) = \nabla^2 A_z(r) = \nabla \cdot \nabla A_z(r) \quad (14)$$

Substitute (14) into the 1st term in (11)

$$\lim_{r_a \rightarrow 0} \left\{ \int_V \nabla \cdot \nabla A_z dV = \oint_{r=r_a} \nabla A_z \cdot d\mathbf{s} \right\} \quad (15)$$

which comes from Divergence theorem $\int_V \nabla \cdot \mathbf{B} dV = \oint \mathbf{B} \cdot d\mathbf{s}$

$$\begin{aligned} \lim_{r_a \rightarrow 0} \left\{ \oint \nabla A_z(r) \big|_{r=r_a} \cdot d\mathbf{s} &= \int_s \left(\frac{d}{dr_a} A_z(r_a) \hat{\mathbf{r}} \right) \cdot \left(r_a^2 \sin \theta d\theta d\phi \hat{\mathbf{r}} \right) \right\} \\ &= r_a^2 \frac{d}{dr_a} A_z(r_a) \int_s \sin \theta d\theta d\phi \\ &= r_a^2 \frac{d}{dr_a} \left(C_1 \frac{e^{-jkr_a}}{r_a} \right) \underbrace{\int_0^{2\pi} \int_0^{\pi} \sin \theta d\theta d\phi}_{=4\pi} \\ &= 4\pi \lim_{r_a \rightarrow 0} \left\{ r_a^2 C_1 \frac{(-jkr_a - 1)}{r_a^2} e^{-jkr_a} \right\} \\ &= -4\pi C_1 \end{aligned} \quad (16)$$

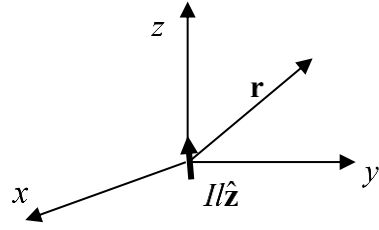
(12), (13) and (16) into (11) gives

$$-4\pi C_1 = -\mu Il$$

$$C_1 = \frac{\mu Il}{4\pi} \quad (17)$$

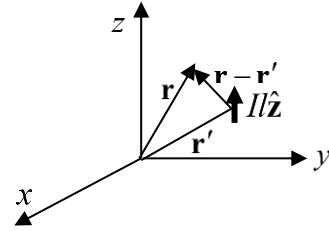
Equation (17) in (10) gives the final result for the potential of a point source, of moment Il , polarized in the z -direction and located at the origin,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu Il}{4\pi} \frac{e^{-jk r}}{r} \hat{\mathbf{z}} \quad (18)$$



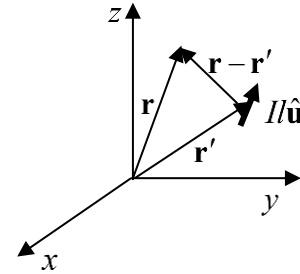
A more general result can be constructed by first moving the source from the origin to some point $\mathbf{r}'$. The field is no longer rotationally symmetric around the origin so $\mathbf{A}$ becomes a function of r, θ and ϕ . This gives

$$\mathbf{A}(\mathbf{r}) = \frac{\mu Il}{4\pi} \frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} \hat{\mathbf{z}} \quad (19)$$



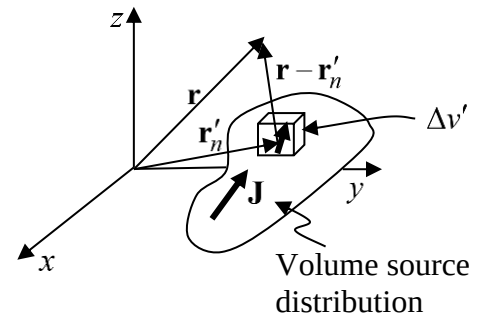
Now tilt the source so that it is in a general direction $\hat{\mathbf{u}}$,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu Il}{4\pi} \frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} \hat{\mathbf{u}} \quad (20)$$



Then for a general source distribution $\mathbf{J}(\mathbf{r})$ in a volume v , $\mathbf{A}(\mathbf{r})$ is obtained by summing all elements with amplitudes $I_n l$, each enclosed in an infinitesimally small volume $\Delta v'$.

$$\begin{aligned} \mathbf{A}(\mathbf{r}) &= \lim_{\Delta v' \rightarrow 0} \sum_n \frac{\mu(I_n l / \Delta v') \hat{\mathbf{u}}_n e^{-jk|\mathbf{r}-\mathbf{r}'_n|}}{4\pi|\mathbf{r}-\mathbf{r}'_n|} \Delta v' \\ &= \int_{v'} \frac{\mu \mathbf{J}(\mathbf{r}') e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|} dv' \end{aligned} \quad (21)$$



$$ds' \begin{array}{c} \uparrow \\ I_n \end{array} \begin{array}{c} \uparrow \\ l \end{array} \quad I_n l / \Delta v' = I_n l / (l \Delta s') = I_n / \Delta s' \xrightarrow{\lim \Delta v' \rightarrow 0} \mathbf{J}_n$$

Magnetic current and charge (set $\mathbf{J} = \rho = 0$)

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$$

$$\nabla \times \mathbf{E} = -\mathbf{M} - j\omega\mu\mathbf{H}$$

$$\nabla \cdot \mathbf{D} = 0$$

$$\nabla \cdot \mathbf{B} = \rho_m$$

Define an electric vector potential $\mathbf{F}$

$$\mathbf{E} = -\frac{1}{\epsilon}\nabla \times \mathbf{F}$$

$$\mathbf{H} = -\frac{j\omega}{k^2}\left(k^2 + \nabla\nabla \cdot\right)\mathbf{F}$$

$$\nabla^2\mathbf{F} + k^2\mathbf{F} = -\epsilon\mathbf{M}$$

$$\nabla \cdot \mathbf{F} = -j\omega\mu\epsilon\Phi_m$$

$$\mathbf{F}(\mathbf{r}) = \int_{V'} \frac{\epsilon\mathbf{M}(\mathbf{r}')e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}dV'$$

If a source contains both electric and magnetic currents then

$$\mathbf{E} = -\frac{j\omega}{k^2}\left(k^2 + \nabla\nabla \cdot\right)\mathbf{A} - \frac{1}{\epsilon}\nabla \times \mathbf{F}$$

$$\mathbf{H} = \frac{1}{\mu}\nabla \times \mathbf{A} - \frac{j\omega}{k^2}\left(k^2 + \nabla\nabla \cdot\right)\mathbf{F}$$

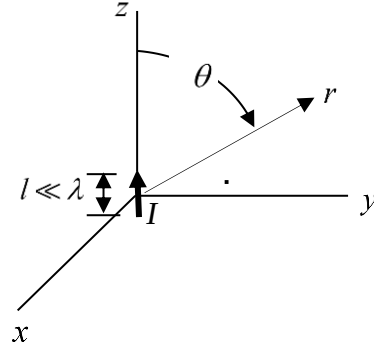
Short dipole $\equiv l \ll \lambda$ Section 2.9

When the current $\mathbf{J}(\mathbf{r})$ is known always start

with
$$\mathbf{A} = \frac{\mu}{4\pi} \int_{v'} \mathbf{J}(\mathbf{r}') \frac{e^{-jkR}}{R} dv'$$

$$\mathbf{J} = I \delta(x') \delta(y') \hat{\mathbf{z}}$$

let $R = |\mathbf{r} - \mathbf{r}'| \doteq r$ because $(r' \ll \lambda)$
and $(r \gg \lambda)$



$$\begin{aligned} \mathbf{A} &= \frac{\mu I}{4\pi} \frac{e^{-jkr}}{r} \int_{v'} \delta(x') \delta(y') dx' dy' dz' \hat{\mathbf{z}} \\ &= \frac{\mu I}{4\pi} \frac{e^{-jkr}}{r} \int_{-l/2}^{l/2} dz' \hat{\mathbf{z}} \\ \mathbf{A} &= \frac{\mu I l e^{-jkr}}{4\pi r} \hat{\mathbf{z}} \end{aligned}$$

change $\hat{\mathbf{z}}$ to spherical coordinates

$$\hat{\mathbf{z}} = \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}$$

so
$$\mathbf{A} = \frac{\mu I l e^{-jkr}}{4\pi r} (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}})$$

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A} = \frac{1}{\mu} \left\{ \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (A_{\phi} \sin \theta) - \frac{\partial A_{\theta}}{\partial \phi} \right] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \phi} A_r - \frac{\partial}{\partial r} (r A_{\phi}) \right] \hat{\boldsymbol{\theta}} \right.$$

Eqns 4-10 in Balanis

$$\left. + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_{\theta}) - \frac{\partial}{\partial \theta} A_r \right] \hat{\boldsymbol{\phi}} \right\}$$

$$\mathbf{H} = \left(-\frac{Ilk^2}{4\pi} \right) \left[\frac{1}{jkr} + \frac{1}{(jkr)^2} \right] \sin \theta e^{-jkr} \hat{\boldsymbol{\phi}}$$

$$\mathbf{E} = -\frac{j\omega}{k^2} (k^2 \mathbf{A} + \nabla \nabla \cdot \mathbf{A}) \quad \text{or} \quad \mathbf{E} = \frac{1}{j\omega \epsilon} \nabla \times \mathbf{H}$$

$$\mathbf{E} = -\frac{Ilk^2 \eta}{4\pi} \left\{ \left[\frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right] 2 \cos \theta e^{-jkr} \hat{\mathbf{r}} + \left[\frac{1}{(jkr)} + \frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right] \sin \theta e^{-jkr} \hat{\boldsymbol{\theta}} \right\}$$

$$\mathbf{E} = -\frac{Ilk^2\eta}{4\pi} \left\{ \left[\frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right] 2 \cos \theta e^{-jkr} \hat{\mathbf{r}} + \left[\frac{1}{(jkr)} + \frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right] \sin \theta e^{-jkr} \hat{\boldsymbol{\theta}} \right\}$$

$$\mathbf{H} = \left(-\frac{Ilk^2}{4\pi} \right) \left[\frac{1}{jkr} + \frac{1}{(jkr)^2} \right] \sin \theta e^{-jkr} \hat{\boldsymbol{\phi}}$$

Homework a) Use $\mathbf{A} = \frac{\mu I l e^{-jkr}}{4\pi r} (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}})$ in $\mathbf{E} = -\frac{j\omega}{k^2} (k^2 + \nabla \nabla \cdot) \mathbf{A}$ to get the expression

$$\mathbf{E} = -\frac{Ilk^2}{4\pi} \eta \left\{ \left[\frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right] 2 \cos \theta e^{-jkr} \hat{\mathbf{r}} + \left[\frac{1}{jkr} + \frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right] \sin \theta e^{-jkr} \hat{\boldsymbol{\theta}} \right\}$$

Throughout your calculation underline all terms associated with the scalar potential, which are all terms that come from $\nabla \nabla \cdot \mathbf{A}$.

(b) You will read in some books that the $\nabla \nabla \cdot \mathbf{A}$ calculation is not necessary. State specifically where in your part (a) calculation the far field would change if $\nabla \nabla \cdot \mathbf{A}$ were not in $\mathbf{E} = -\frac{j\omega}{k^2} (k^2 + \nabla \nabla \cdot) \mathbf{A}$.

Notice the far field can also be found by replacing $\nabla \rightarrow -jk\hat{\mathbf{r}}$. This trick is useful, particularly when the current and therefore the potential is a direction other than the z-direction. Basically the $\nabla \nabla \cdot = (-jk\hat{\mathbf{r}})(-jk\hat{\mathbf{r}}) = -k^2 \hat{\mathbf{r}} \hat{\mathbf{r}} \cdot$ term exists to remove the radial component of the far field. This trick is on pg 133 in Harrington.

$$\begin{aligned} \mathbf{E} = -\frac{j\omega}{k^2} (k^2 + \nabla \nabla \cdot) \mathbf{A} - \frac{1}{\epsilon} \nabla \times \mathbf{F} & \xrightarrow{\text{for far field}} \mathbf{E} = -\frac{j\omega}{k^2} (k^2 - k^2 \hat{\mathbf{r}} \hat{\mathbf{r}} \cdot) \mathbf{A} - \left(\frac{-jk}{\epsilon} \hat{\mathbf{r}} \right) \times \mathbf{F} \\ & = -j\omega (1 - \hat{\mathbf{r}} \hat{\mathbf{r}} \cdot) \mathbf{A} + \frac{jk}{\epsilon} \hat{\mathbf{r}} \times \mathbf{F} \end{aligned}$$

Method of gradients (author: C. T. Tai) : Express the unit vectors in one coordinate system in terms of the unit vectors in another coordinate system.

Knowing $\nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{x}} + \frac{\partial f}{\partial y} \hat{\mathbf{y}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}} = \frac{\partial f}{\partial \rho} \hat{\boldsymbol{\rho}} + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \hat{\boldsymbol{\phi}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}}$

and $x = \rho \cos \phi, y = \rho \sin \phi, z = z$

we can find $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ in terms of $\hat{\boldsymbol{\rho}}, \hat{\boldsymbol{\phi}}, \hat{\mathbf{z}}$

$$\nabla x = \hat{\mathbf{x}} = \frac{\partial}{\partial \rho}(\rho \cos \phi) \hat{\boldsymbol{\rho}} + \frac{1}{\rho} \frac{\partial}{\partial \phi}(\rho \cos \phi) \hat{\boldsymbol{\phi}} + \frac{\partial}{\partial z}(\rho \cos \phi) \hat{\mathbf{z}}$$

$$\hat{\mathbf{x}} = \cos \phi \hat{\boldsymbol{\rho}} - \sin \phi \hat{\boldsymbol{\phi}}$$

$$\nabla y = \hat{\mathbf{y}} = \frac{\partial}{\partial \rho}(\rho \sin \phi) \hat{\boldsymbol{\rho}} + \frac{1}{\rho} \frac{\partial}{\partial \phi}(\rho \sin \phi) \hat{\boldsymbol{\phi}} + \frac{\partial}{\partial z}(\rho \sin \phi) \hat{\mathbf{z}}$$

$$\hat{\mathbf{y}} = \sin \phi \hat{\boldsymbol{\rho}} + \cos \phi \hat{\boldsymbol{\phi}}$$

$$\nabla z = \hat{\mathbf{z}} = \frac{\partial}{\partial \rho}(z) \hat{\boldsymbol{\rho}} + \frac{1}{\rho} \frac{\partial}{\partial \phi}(z) \hat{\boldsymbol{\phi}} + \frac{\partial}{\partial z}(z) \hat{\mathbf{z}}$$

$$\hat{\mathbf{z}} = \hat{\mathbf{z}}$$

Or we can find $\hat{\boldsymbol{\rho}}, \hat{\boldsymbol{\phi}}, \hat{\mathbf{z}}$ in terms of $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$

$$\hat{\boldsymbol{\rho}} = (\hat{\boldsymbol{\rho}} \cdot \hat{\mathbf{x}}) \hat{\mathbf{x}} + (\hat{\boldsymbol{\rho}} \cdot \hat{\mathbf{y}}) \hat{\mathbf{y}} + (\hat{\boldsymbol{\rho}} \cdot \hat{\mathbf{z}}) \hat{\mathbf{z}}$$

$$=0$$

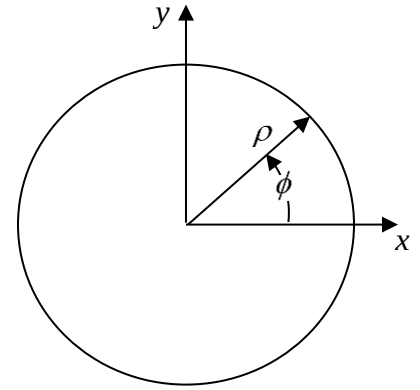
$$= \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}$$

$$\hat{\boldsymbol{\phi}} = (\hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{x}}) \hat{\mathbf{x}} + (\hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{y}}) \hat{\mathbf{y}} + (\hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{z}}) \hat{\mathbf{z}}$$

$$=0$$

$$= -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}$$

$$\hat{\mathbf{z}} = (\hat{\mathbf{z}} \cdot \hat{\mathbf{x}}) \hat{\mathbf{x}} + (\hat{\mathbf{z}} \cdot \hat{\mathbf{y}}) \hat{\mathbf{y}} + (\hat{\mathbf{z}} \cdot \hat{\mathbf{z}}) \hat{\mathbf{z}} = \hat{\mathbf{z}}$$



Spherical Coordinates

Know: $x = r \sin \theta \cos \phi$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

and

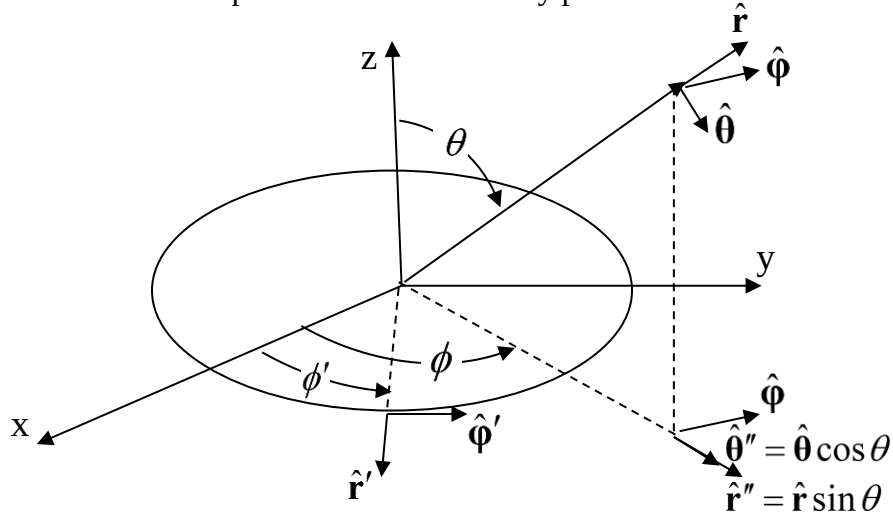
$$\nabla f = \frac{\partial f}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\boldsymbol{\phi}}$$

Homework: Using the method of gradients

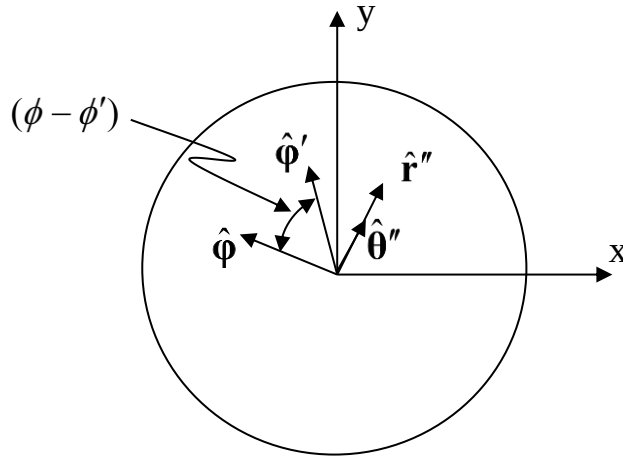
- find $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ in terms of spherical coordinates r, θ, ϕ and unit vectors $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$
- find $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$ in terms of spherical coordinates r, θ, ϕ and Cartesian unit vectors.
- find $|\mathbf{r} - \mathbf{r}'|$ in spherical coordinates. Do not approximate any steps.
- find $|\mathbf{r} - \mathbf{r}'|$ in cylindrical coordinates. Do not approximate any steps.

- Geometrical method for expressing prime (source) coordinate unit vectors in terms of unprimed (observation point) coordinates in variant (cylindrical and spherical) coordinate systems.
- *Graphical projection method:* The vector $\hat{\phi}'$ is a function of ϕ' , so it must be expressed in terms of the observation point vectors $\hat{r}$, $\hat{\theta}$, and $\hat{\phi}$. These vectors are fixed at a point in space and therefore are not effected by the integration in ϕ' .

First project the observation point vectors into the x-y plane.



To get a clearer picture, maintain the direction of the vectors but move them into the origin.



Top View

Now project the projected (double primed) observation point vectors into the $\hat{\phi}'$ direction. The result is:

$$\begin{aligned}\hat{\phi}' &= (\hat{r}'' \cdot \hat{\phi}')\hat{r}' + (\hat{\theta}'' \cdot \hat{\phi}')\hat{\theta}' + (\hat{\phi} \cdot \hat{\phi}')\hat{\phi} \\ &= \sin \theta \sin(\phi - \phi')\hat{r}' + \cos \theta \sin(\phi - \phi')\hat{\theta}' + \cos(\phi - \phi')\hat{\phi}\end{aligned}$$

- Mathematical method for expressing prime (source) coordinate unit vectors in terms of unprimed (observation point) coordinates in variant (cylindrical and spherical) coordinate systems.

For example

$$\begin{aligned}
 \hat{\phi}' &= (\hat{\phi}' \cdot \hat{r})\hat{r} + (\hat{\phi}' \cdot \hat{\theta})\hat{\theta} + (\hat{\phi}' \cdot \hat{\phi})\hat{\phi} \\
 &= [(-\sin \phi' \hat{x} + \cos \phi' \hat{y}) \cdot (\sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z})]\hat{r} \\
 &\quad + [(-\sin \phi' \hat{x} + \cos \phi' \hat{y}) \cdot (\cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z})]\hat{\theta} \\
 &\quad + [(-\sin \phi' \hat{x} + \cos \phi' \hat{y}) \cdot (-\sin \phi \hat{x} + \cos \phi \hat{y})]\hat{\phi} \\
 &= [-\sin \phi' \sin \theta \cos \phi + \cos \phi' \sin \theta \sin \phi]\hat{r} \\
 &\quad + [-\sin \phi' \cos \theta \cos \phi + \cos \phi' \cos \theta \sin \phi]\hat{\theta} \\
 &\quad + [\sin \phi' \sin \phi + \cos \phi' \cos \phi]\hat{\phi} \\
 &= \sin \theta \sin(\phi - \phi')\hat{r} + \cos \theta \sin(\phi - \phi')\hat{\theta} + \cos(\phi - \phi')\hat{\phi}
 \end{aligned}$$

